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Predicting Judicial Decisions: A Statistically Rigorous Approach and a New Ensemble Classifier

Andrea Visentin, Alessia Nardotto, Barry O'Sullivan

Insight

SFI RESEARCH CENTRE FOR DATA ANALYTICS

Thanks to Dr Bridge for part of the lecture material

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Free-Form Text

- We've looked at AI systems that can handle structured data (i.e. numeric-valued and non-numeric-valued features).
- Suppose instead the objects in your dataset are **documents**, rather than houses, students, irises or photos.
  - E.g. web pages, tweets, blog posts, emails, posts to Internet forums and chatrooms, ...
  - They might have a little structure to them (headings and so on), but they are primarily **free-form text**, written in a **natural language** such as English.

Free-Form Text

- Many AI algorithms can only handle vectors of numbers. So one way to apply AI techniques to a dataset of documents is to convert the raw text in the documents into vectors of numbers.
- **Note that in this lecture, each document becomes one vector.** In the next lecture, each word becomes one vector and hence a document becomes a list of vector

EUROPEAN COURT OF HUMAN RIGHTS

COUR EUROPÉENNE DES DROITS DE L'HOMME

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European Court of Human Rights

The ***European Court of Human Rights*** (ECHR) is a supranational or international court of the Council of Europe which interprets the ***European Convention on Human Rights***. The court hears applications alleging that a contracting state has breached one or more of the human rights enumerated in the Convention or its optional protocols to which a member state is a party. (Wikipedia)

European Convention on Human Rights

- Life
- Freedom of expression
- Torture
- Fair Trial
- Discrimination
- Privacy
- Liberty and Security
- And much more

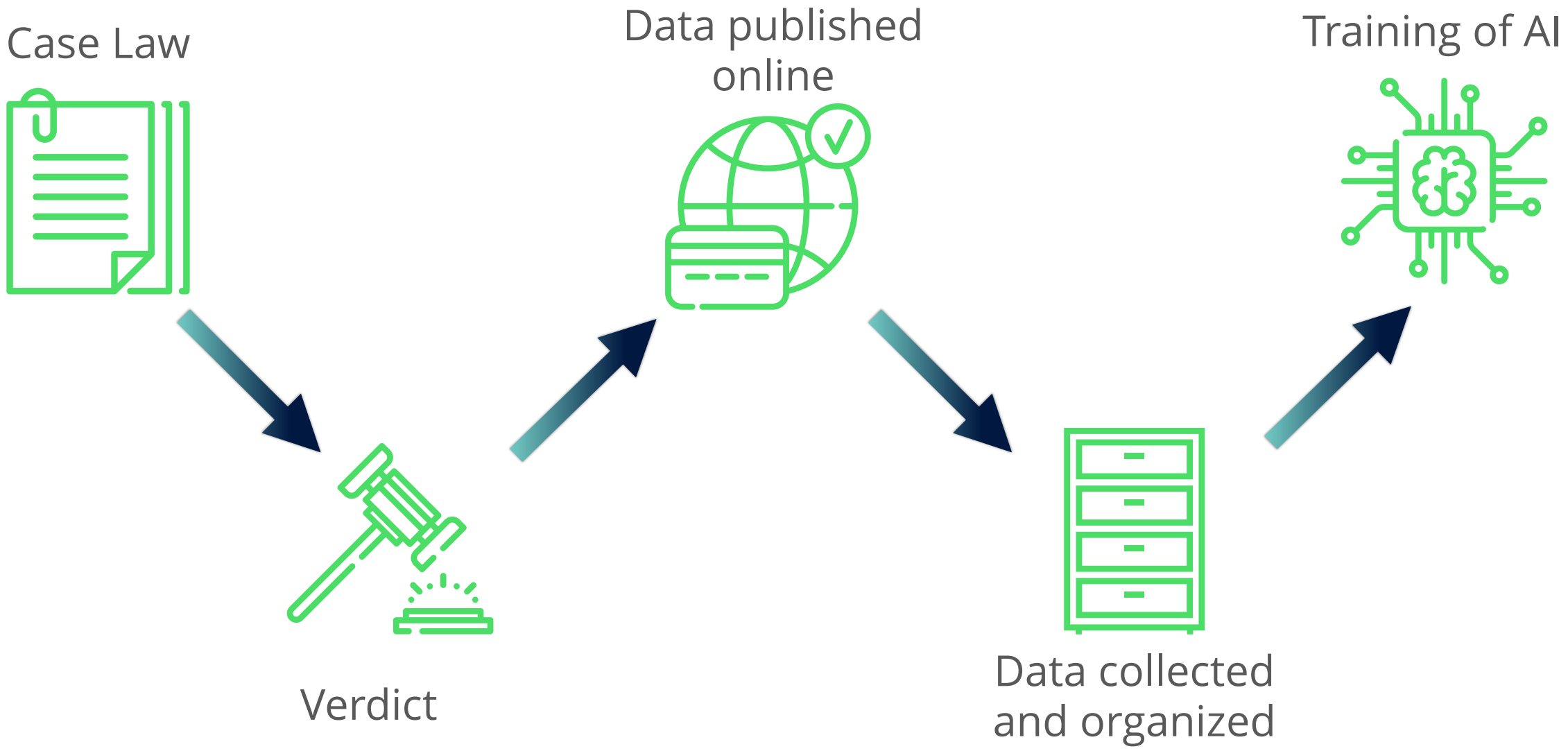
Possible Violation

Case Law

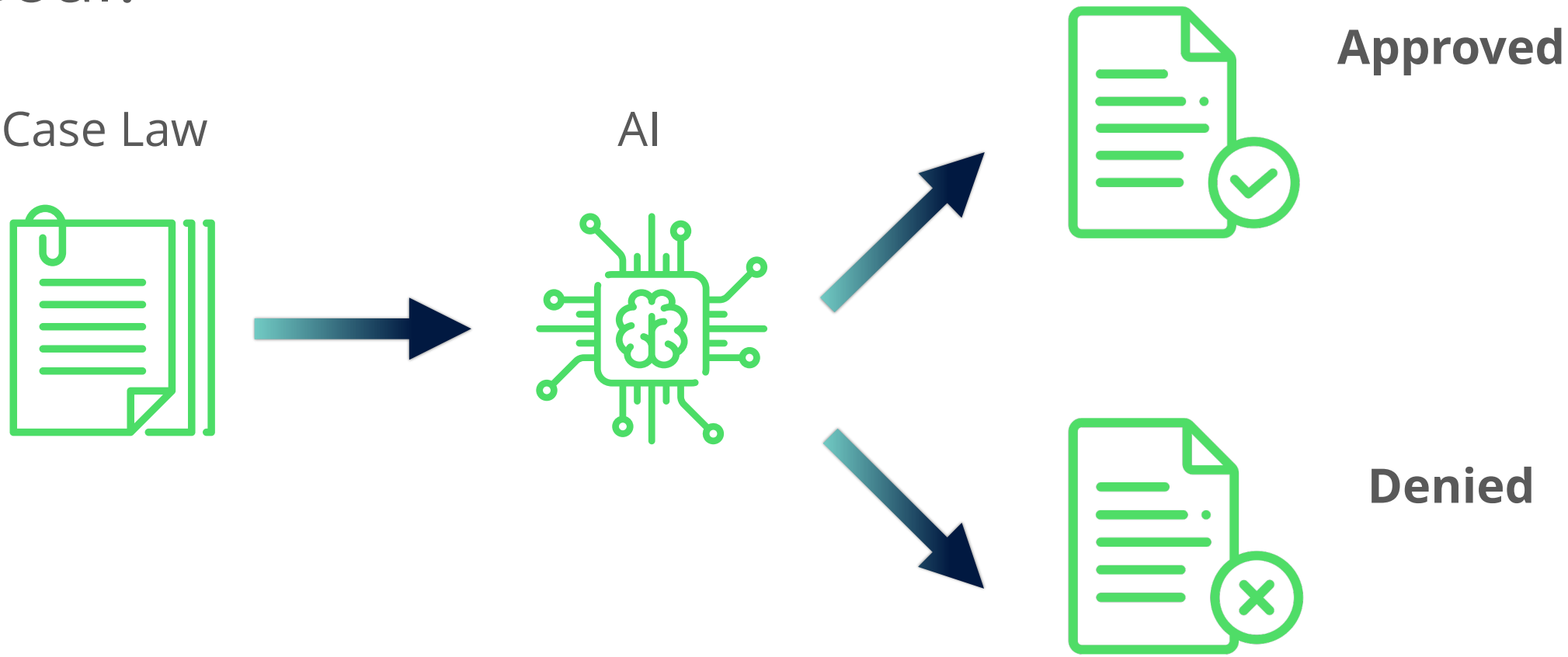
Verdict

Approved
There is a violation

Denied
There is not a violation


Case Law
Verdict
Data published online
Data collected and organized
Training of AI

Goal?



What is the name of this task?

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Original work

N. Aletras, D. Tsarapatsanis, D. Preotiuc-Pietro, and V. Lampos, "Predicting judicial decisions of the European Court of Human Rights: a Natural Language Processing perspective"

Table 1 **Articles of the Convention and number of cases in the data set.** Article numbers, Convention right that each article protects and the number of cases in our data set.

Article	Human Right	Cases
3	Prohibits torture and inhuman and degrading treatment	250
6	Protects the right to a fair trial	80
8	Provides a right to respect for one's "private and family life, his home and his correspondence"	254

Natural Language Processing

Natural language processing (NLP) is a subfield of linguistics, computer science, and artificial intelligence concerned with the interactions between computers and human language, in particular how to program computers to process and analyze large amounts of natural language data. (Wikipedia)

Components of the system

Preprocessing of the data

Cleaning
Stop words
Normalization

Features extraction

Bag of Words
N-grams
Topics

Classifier training and testing

10 runs 10-cv
SVM

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Cleaning

- Removing punctuation.
- Removing names, dates, places.
- Removing any indication of the final decision.
- More?

Stop words

Stop words usually refers to the most common words in a language. There is no universal list of ***stop words*** that is used by all NLP tools in common.

E.g.:

- The
- Of
- A
- In
- For

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Feature Extraction

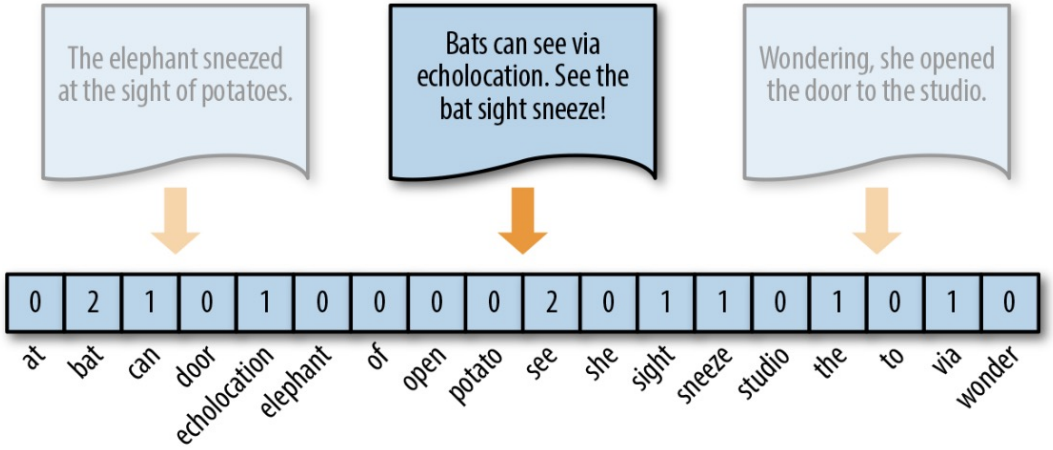
Feature extraction is a general term for methods of constructing combinations of the variables to get around these problems while still describing the data with sufficient accuracy.

- Extract numerical or categorical features from the data.
- These features have to represent the data.

Bag of Words

Feature extraction technique that transforms a text into a list of words occurrences.

It simply counts how many times a word appears!



Bag of Words

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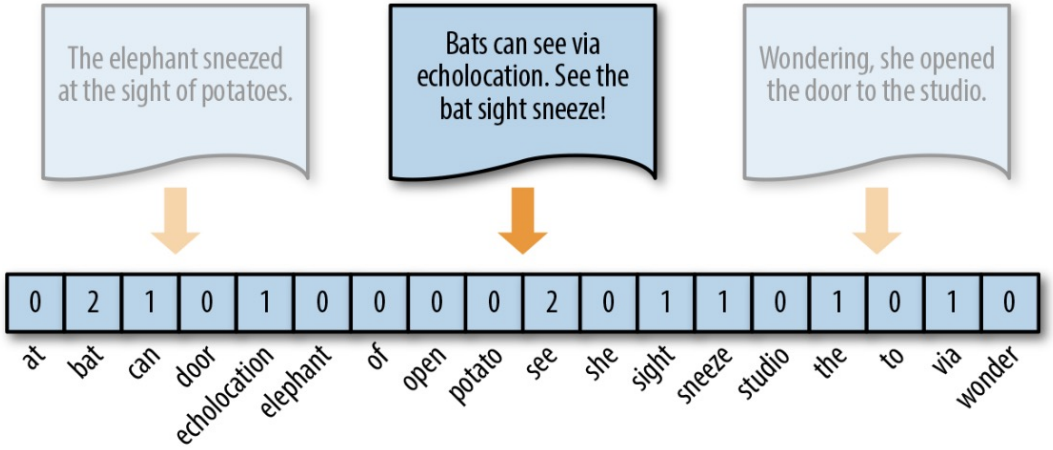

N-gram

The **N-gram** is a generalization of the Bag of Words. It counts groups of consecutive words.

The witness saw it clearly!

- 2-grams: “The witness”, ”witness saw”, “saw it”, “it clearly”.
- 3-grams: “The witness saw”, ”witness saw it”, “saw it clearly”.
- 4-grams: “The witness saw it”, ”witness saw it clearly”.

Bag of Words - Stop words

Which are the limits?

Possible drawbacks?

Bag of Words - Stop words

- Russia attacked Ukraine
- Russia was attacked by Ukraine
- Ukraine attacked Russia

All have the same BoW after the stop words

{'Russia', 'Attacked', 'Ukraine'} but the meaning is quite different

Bag of Words - Stop words

- It loses all the information that English conveys through the order of words in sentences,
  - e.g. "People learn to hate" and "People hate to learn" have very different meanings but end up with the same bag-of-words representation.
- It loses the information that English conveys using its stop-words, most notably negation,
  - e.g. "They hate religion" and "I do not hate religion" will have the same bag-of-words representation.
- This may not matter for some applications (e.g. spam detection) but will matter for others (e.g. machine translation), for which you need a different representation.

Topics

N-grams that are semantically similar are clustered. These vectors represents the 30 most frequent topics in the court cases.

Topic	Label	Words
		Top-5 Violation
4	Positive State Obligations	injury, protection, ordered, damage, civil, caused, failed, claim, course, connection, region, effective, quashed, claimed, suffered, suspended, carry, compensation, pecuniary, ukraine
10	Detention conditions	prison, detainee, visit, well, regard, cpt, access, food, situation, problem, remained, living, support, visited, establishment, standard, admissibility merit, overcrowding, contact, good
3	Treatment by state officials	police, officer, treatment, police officer, July, ill, force, evidence, ill treatment, arrest, allegation, police station, subjected, arrested, brought, subsequently, allegedly, ten, treated, beaten

Final result

	"Jury"	"ill"	"police station"	"no respect"	Topic 1	Topic 2	Label
Case 1	20	3	0	2	15	23	Violation
Case 2	5	0	14	0	4	14	Non-violation
Case 3	2	5	2	1	18	12	Violation
Case 4	7	9	3	0	31	9	Violation

Final result

	"Jury"	"ill"	"police station"	"no respect"	Topic 1	Topic 2	Label
Case 1	20	3	0	2	15	23	Violation
Case 2	5	0	14	0	4	14	Non-violation
Case 3	2	5	2	1	18	12	Violation
Case 4	7	9	3	0	31	9	Violation

What is normalization?

Normalization

	"Jury"	"ill"	"police station"	"no respect"	Topic 1	Topic 2	Label
Case 1	20/20	3/9	0/14	2/2	15/31	23/23	Violation
Case 2	5/20	0/9	14/14	0/2	4/31	14/23	Non-violation
Case 3	2/20	5/9	2/14	1/2	18/31	12/23	Violation
Case 4	7/20	9/9	3/14	0/2	31/31	9/23	Violation

Why we do it?

Normalization v2

	"Jury"	"ill"	"police station"	"no respect"	Topic 1	Topic 2	Label
Case 1	20/23	3/23	0/23	2/23	15/23	23/23	Violation
Case 2	5/14	0/14	14/14	0/14	4/14	14/14	Non-violation
Case 3	2/18	5/18	2/18	1/18	18/18	12/18	Violation
Case 4	7/31	9/31	3/31	0/31	31/31	9/31	Violation

Why we do it?

Feature Selection

The cases are divided in 4 parts of the document:

- Procedure
- Circumstances
- Relevant Law
- Law
- Facts

We consider them independently and all together.

Components of the system

Preprocessing of the data

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Stop words

Features extraction

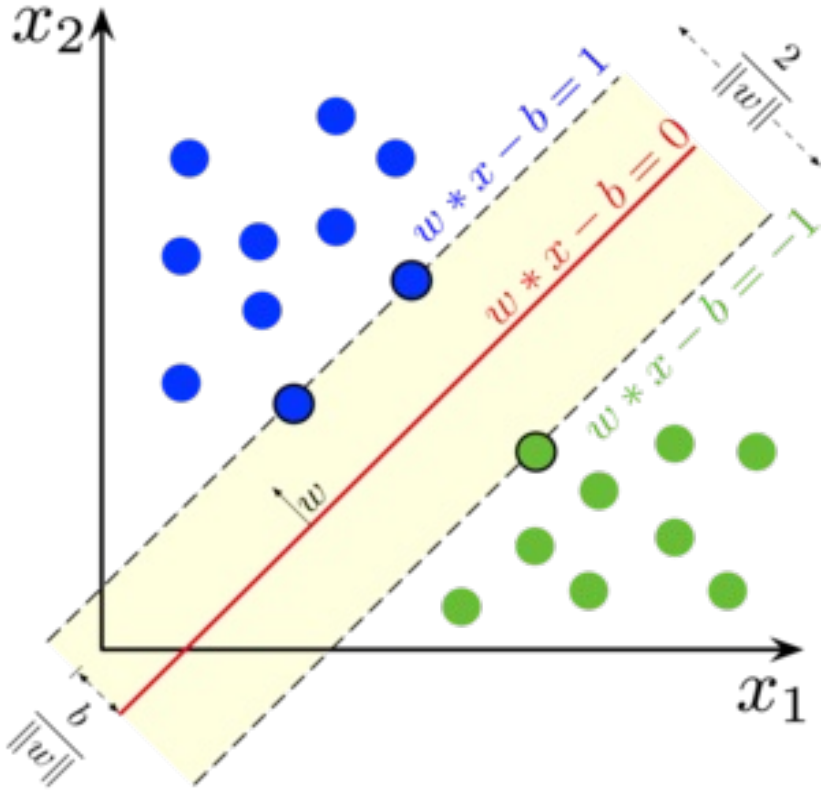
Bag of Words
N-grams
Topics

Classifier training and testing

10 runs 10-cv
SVM

Classifier - SVM

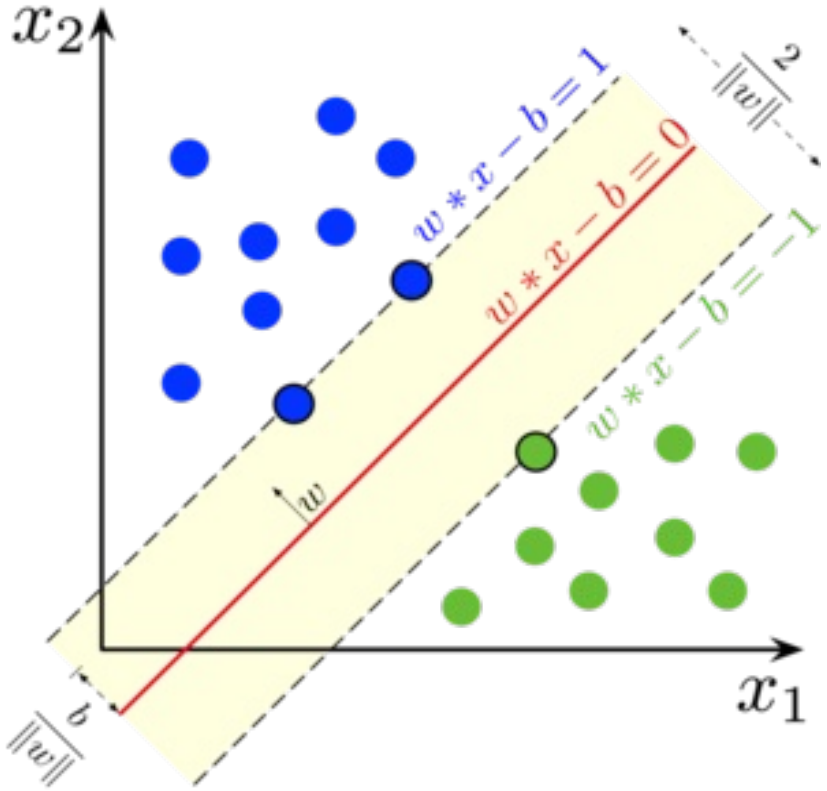
Support Vector Machines



Classifier - kNN

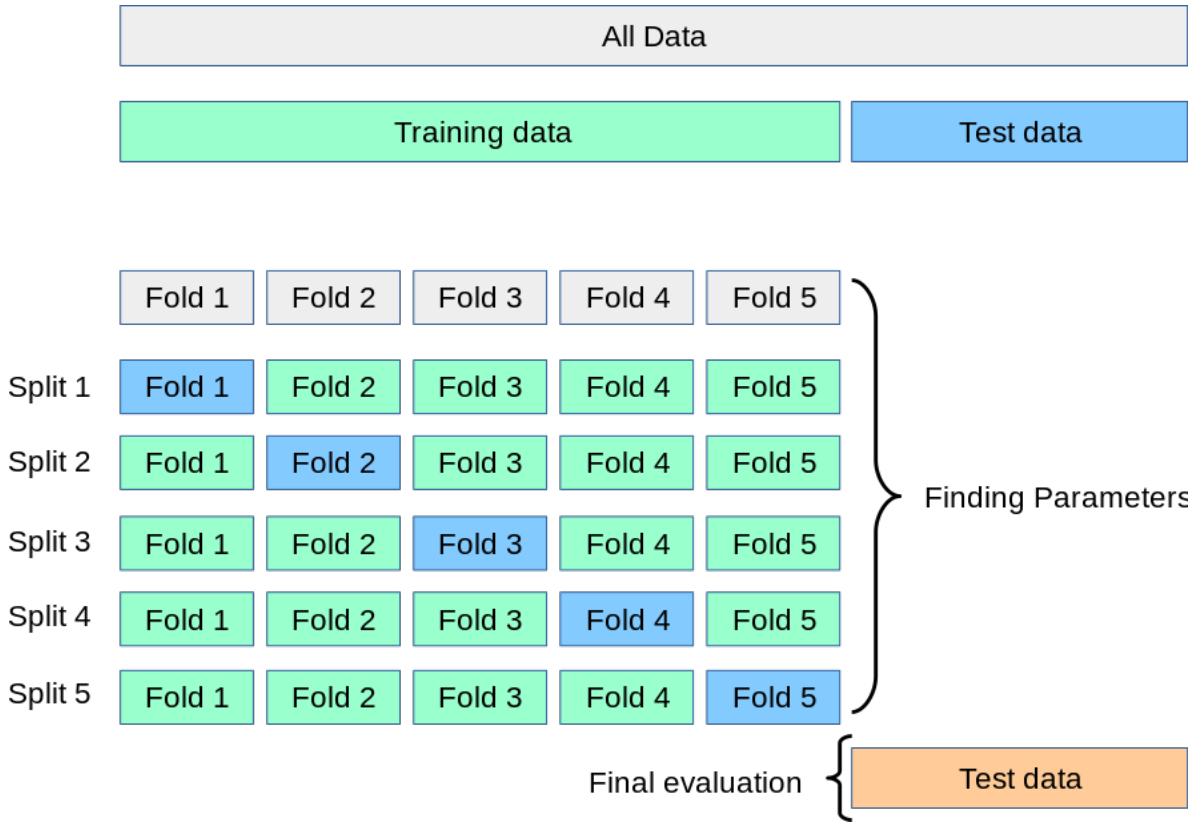
The same can be done by using

kNN or Decision Trees



Training – Testing approach

10-Cross Validation

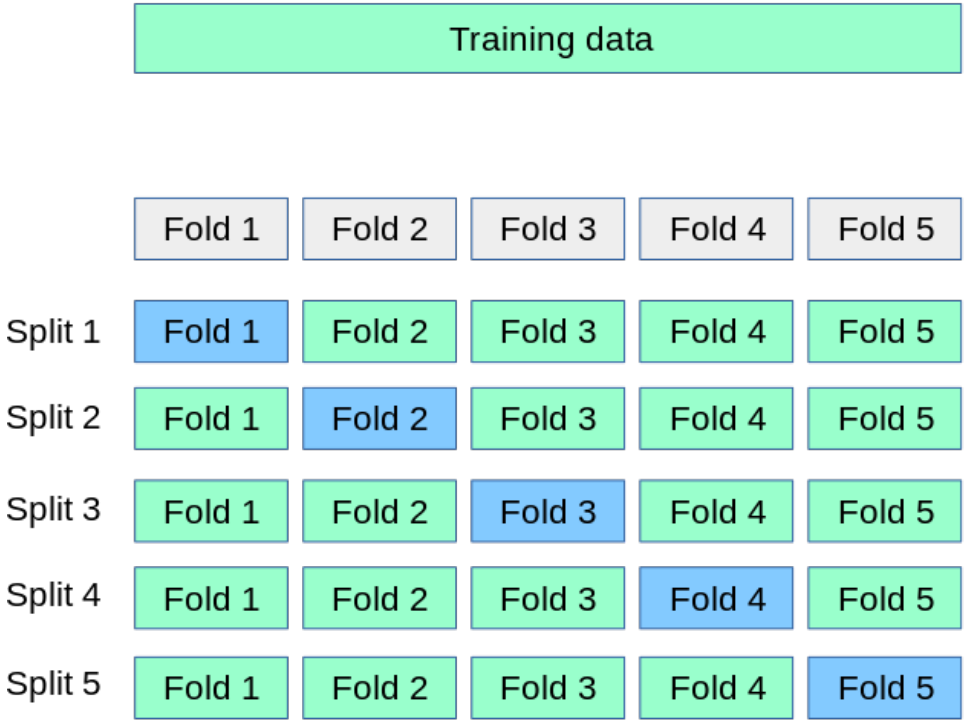


Training – Testing approach

10-Cross Validation.

For small dataset we can apply it over the whole dataset.

Over 10 **random** shuffles!! The right shuffle can give a **fake** improvement of up to 9% in this case.



Results

Feature Type	Article 3	Article 6	Article 8	Average
Full	70.3	84.2	76.1	76.9
Procedure	65.6	76.6	65.0	69.1
Circumstances	71.3	82.9	75.9	76.7
Relevant Law	70.6	75.6	73.7	73.3
Facts	71.6	81.9	75.1	76.2
Law	48.3	67.9	61.1	59.1
Topics	**75.5**	81.5	76.2	77.7
Topics and Cir	73.8	81.1	77.6	77.5

Visentin, Andrea, Alessia Nardotto, and Barry O'Sullivan. "Predicting Judicial Decisions: A Statistically Rigorous Approach and a New Ensemble Classifier." 2019 IEEE 31st International Conference on Tools with Artificial Intelligence (ICTAI). IEEE, 2019.

Suggested read

Medvedeva, Masha, Michel Vols, and Martijn Wieling. "Using machine learning to predict decisions of the European Court of Human Rights." Artificial Intelligence and Law 28.2 (2020): 237-266.

What is better than a robot judge?

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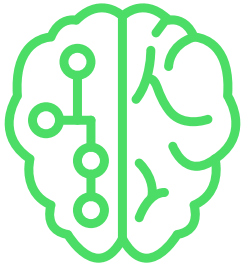
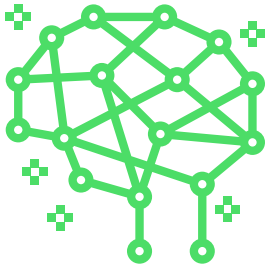
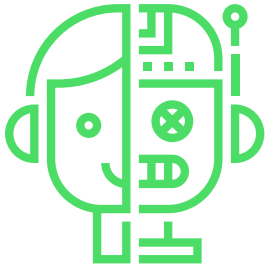
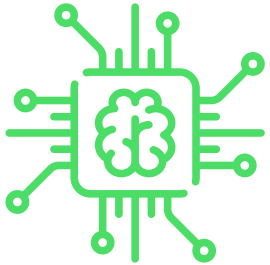
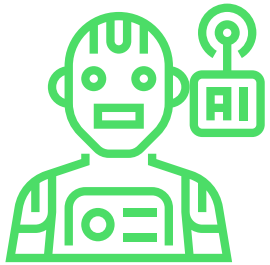
Judicial Panel











Our contribution

Extension of Aletras et al work.

- Bayesian statistically sound method to compare classifier.
- A new way to make juridical decisions based on documents with different subsections, motivated by work from ensemble classifiers.
- An extension of the previous technique to multiple classifiers.

Voting classifier

- The cases are divided in 5 subsections: *Procedure, Circumstances, Relevant Law, Facts, Law*.
- Plus we have the *Full* document and the clustered *Topics.*
- We trained SVMs on the different features and we aggregate their "votes".

Voting classifier

	Procedure	Circumstances	Relevant Law	Law	Facts	Full	Topics	**Final Vote**
Case 1	Yes	No	Yes	Yes	No	Yes	Yes	**Yes (5/7)**
Case 2	Yes	No	Yes	No	Yes	No	No	**No (4/7)**
Case 3	No	No	No	No	No	No	No	**No (7/7)**
Case 4	Yes	No	Yes	No	Yes	Yes	No	**Yes (4/7)**

Violation Yes/No

Results

Feature Type	Article 3	Article 6	Article 8	Average
Full	70.3	84.2	76.1	76.9
Procedure	65.6	76.6	65.0	69.1
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Law	48.3	67.9	61.1	59.1
Topics	**75.5**	81.5	76.2	77.7
Topics and Cir	73.8	81.1	**77.6**	77.5
Voting	72.9	**86.1**	77.4	**78.8**

Components of the system

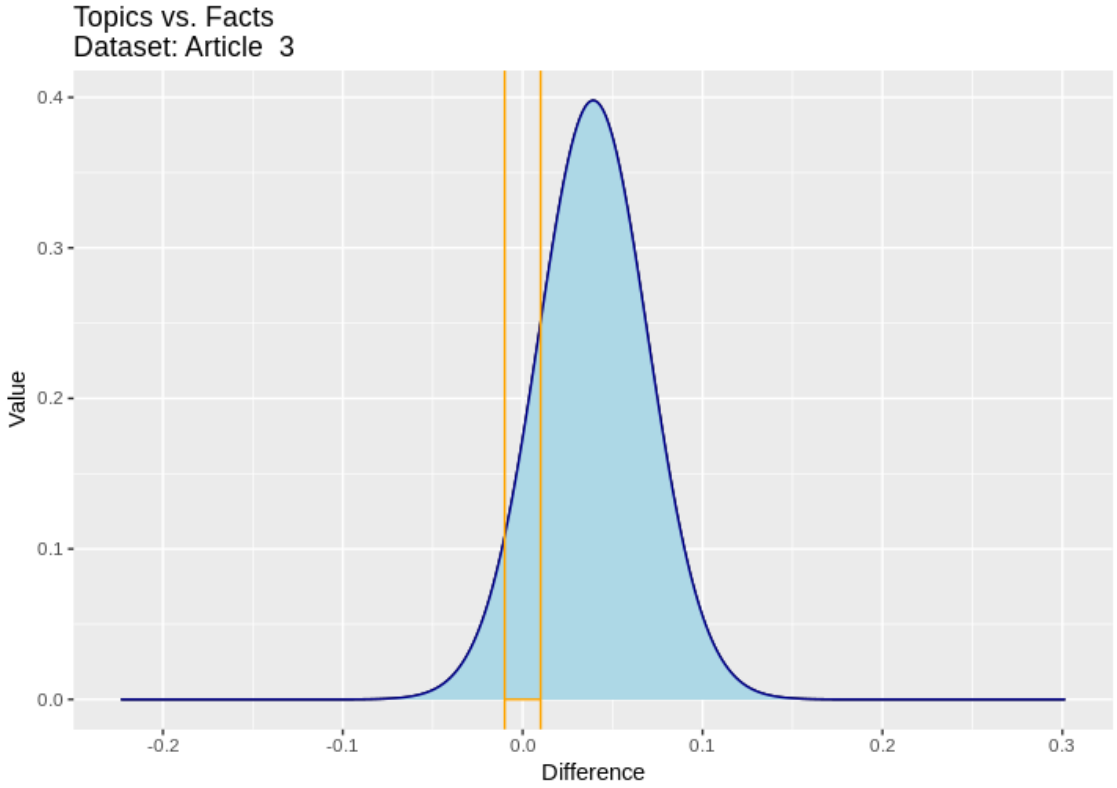
Changing the preprocessing and the feature selection we created a variety of solutions.

Classifier comparison

When the difference between algorithms is ***statistically significant***?

What are the chances that the observed difference is random and one isn't better than the other?

Bayesian Classifier Comparison

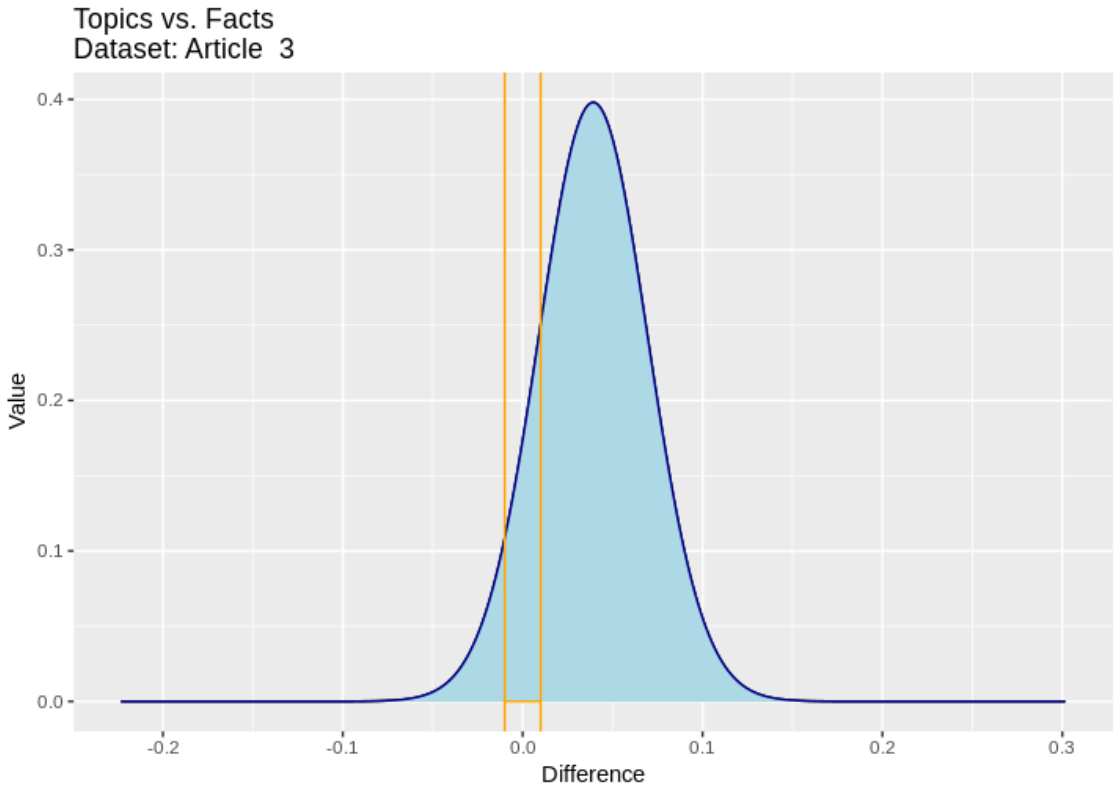


[2] A. Benavoli, G. Corani,J. Demsar, and M. Zaffalon, “Time for a change: a tutorial for comparing multiple classifiers through Bayesian analysis”

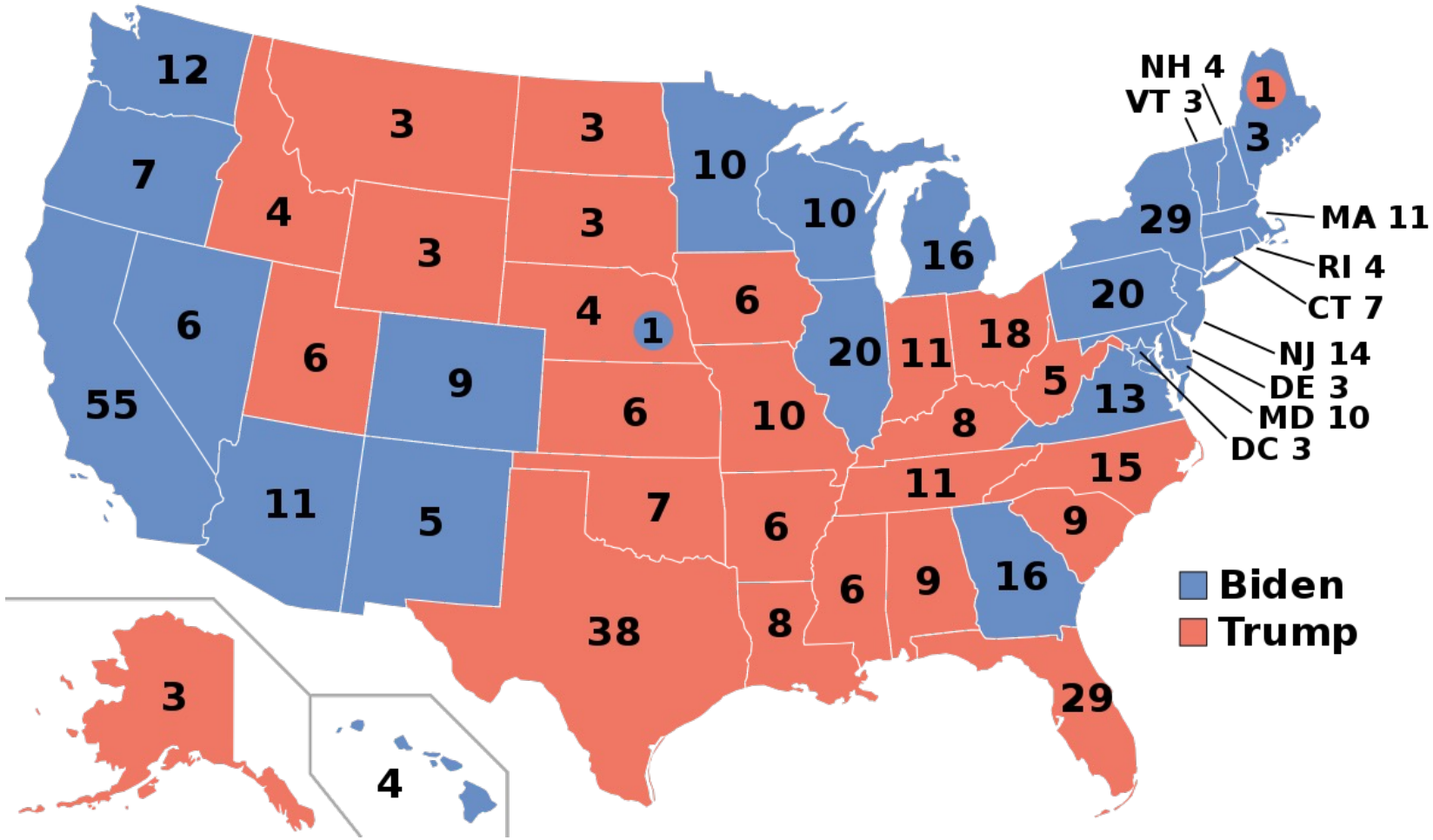
Computes three probabilities

- $P_{A>>B}$: posterior probability of the mean difference of accuracies being practically in favour of Algorithm A
- $P_{A==B}$ or P_{rope} : the posterior probability of the two classifiers being practically equivalent
- $P_{A<<B}$: posterior probability of the mean difference of accuracies being in favour of Algorithm B

Bayesian Classifier Comparison



What is better than a robot judicial panel?



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Components of the system

We can consider different classifiers! Like?

Ensemble of ensembles

- We trained different classifiers on the same voting scheme presented before.
- We used: *SVM, GMB* and *KNN.*
- We applied the majority vote.

Ensemble of ensembles

Classifier	Article 3	Article 6	Article 8	Average
SVM	72.9	**86.1**	77.4	78.8
GBM	73.8	83.8	75.1	77.6
KNN	69.6	80.1	77.1	75.6
Ensemble	**73.9**	85.1	**78.0**	**79.0**

Why we can’t reach 100%?

Some answers

- Neither another human judge will agree with 100% of the decisions.
- We lose information. Like?

Is this system safe?
How can I trick it?

Some answers

- I discover that a particular word strongly influence the AI decision (e.g. “police”). I add many occurrences of it in the text:
  - The policeman was in the police station. He took the police car. He identified himself as member of the police.
- I copy a case that previously got accepted.

Possible applications

Understand the reasons of the outcome

Support judges decisions

Help lawyers to make a stronger case

Thanks for the attention

Questions?